Machine Learning for Predicting Sediment Particle Size Distributions

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1. Problem Statement

Marine sediment particles are important in environmental and sediment-transport processes. Their size can change rapidly because of aggregation and disaggregation (flocculation), making particle-size prediction difficult using fixed empirical relationships. The reference study by Galen Egan investigates whether machine-learning regression can predict properties of suspended sediment particle size distributions (PSDs). The two prediction targets are the median particle diameter (d50) and the variance of the PSD (σ^2). The objective of this mini-project is to reproduce the Random Forest (RF) and Support Vector Regression (SVR) methodology from the reference and extend the comparison using XGBoost regression.

2. Dataset Details

The reference dataset was collected during three one-month field campaigns in South San Francisco Bay: July–August 2018, January–February 2019, and April–May 2019. Suspended sediment PSDs were measured hourly using a Laser In-Situ Scattering and Transmissometry meter (LISST-100x). The reference reports 1,648 examples at the selected instrument platform. For this project, the closest publicly available/reconstructed version of the described dataset was used; therefore, the local results are treated as a reproduction experiment rather than an exact copy of the author's original processed machine-learning table.

Category	Variables	Role / Unit
Hydrodynamic	ub, u	Wave-induced and mean tidal velocity
Water chemistry	S, T	Salinity and temperature
Optical / biological	np, chl-a, a676/a650, a450/a676	Environmental/optical predictors
Targets	d50, σ^2	Particle diameter (μm) and PSD variance (μm^2)

The project dataset contains 1,648 observations, eight input variables, and two continuous targets. Missing values were checked before model training.

3. Approach

Because d50 and σ^2 are continuous quantities, the task is formulated as a regression problem. Following the reference, Random Forest feature importance was used to identify the most useful inputs. The reference averaged feature importance over 1,000 RF tests using 10 estimators and a 70/30 train-test split. The four selected features for subsequent modelling are salinity (S), near-bottom wave-induced velocity (ub), particle index of refraction (np), and temperature (T).

4. Reference Models

Random Forest regression was implemented with 32 trees and a 70/30 train-test split, matching the reference final setting. The reference selected approximately 32 trees after the out-of-bag (OOB) score became relatively flat. Support Vector Regression used a 70% training, 15% cross-validation, and 15% test split, with C = 2048 and ε = 4. Model quality was measured using the coefficient of determination (R^2).

5. Implementation Overview

The implementation was developed in Python using pandas, NumPy, scikit-learn, Matplotlib, and XGBoost. The workflow is: load the CSV dataset → inspect shape, columns and missing values → select the reference features → split the data → train RF and SVR → generate predictions → calculate R^2 for d50 and σ^2 → train the XGBoost extension → compare model results. The Colab notebook also generates feature-importance and actual-versus-predicted plots and exports the model comparison and feature-importance results as CSV files.

6. Reproduction Results and Extension

The table compares the reference paper's reported R^2 values with the results currently recorded for this project. XGBoost is the extension model and was not used in the reference paper, so it has no reference score.

Model	Target	Reference R^2	Our R^2
Random Forest	d50	0.81	0.900
Random Forest	σ^2	0.75	0.764
SVR	d50	0.76	0.885
SVR	σ^2	0.76	0.747
XGBoost (extension)	d50	N/A	0.899
XGBoost (extension)	σ^2	N/A	0.762

Interpretation: Random Forest gives the highest project R^2 for both targets (0.900 for d50 and 0.764 for σ^2). XGBoost performs very similarly (0.899 and 0.762), making it a competitive additional model on this dataset. SVR achieves 0.885 for d50 and 0.747 for σ^2.

7. Project Extension — XGBoost

XGBoost Regression was added as the project extension because it was not one of the two reference models. It uses the same four selected features and the same regression targets. The implementation uses a reproducible configuration with 100 estimators, maximum depth 3, learning rate 0.05, subsample 0.8, and column subsampling 0.8. Performance is evaluated using R^2 so that the extension can be compared with RF and SVR under the same metric.

8. Conclusions

This mini-project demonstrates an end-to-end machine-learning approach for predicting sediment particle-size properties from environmental and hydrodynamic measurements. The work reproduces the main modelling methodology of the reference study using Random Forest and SVR and extends it with XGBoost. Feature selection reduces the original eight predictors to four reference-selected variables: S, ub, np, and T. The obtained R^2 values show useful regression performance on the project dataset, while the differences from the published values are documented rather than treated as an exact reproduction. The project also provides a reproducible Colab notebook and saved result files for demonstration.

Reference

Galen Egan, “Machine learning for predicting sediment particle size distributions,” supplied reference report for the mini-project.